INTERNSHIP PROJECT REPORT

EDXcellence
Learn without Limits >>>

ARTIFICIAL INTELLIGENCE WITH PYTHON

DURATION - 3 MONTHS

SUBMITTED BY

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MINOR PROJECT

SPOTIFY DATA ANALYTICS & SONG RECOMMENDATION ENGINE

1. INTRODUCTION

With the exponential growth of digital music platforms like Spotify, users are exposed to millions of songs, making music discovery a complex problem. Recommendation systems play a crucial role in enhancing user experience by filtering relevant content.

This project presents a data-driven music analytics and recommendation system that leverages -

- Exploratory Data Analysis (EDA).
- Unsupervised Machine Learning.
- Content-Based Filtering.

The system is implemented as a full-stack interactive dashboard using Streamlit, allowing users to explore music data and receive personalized song recommendations.

2. PROBLEM STATEMENT

Users often struggle to discover songs that match their preferences due to -

- Massive content volume.
- Lack of structured similarity insights.
- Limited explainability in recommendations.

Solution -
Design a system that -

- Understands musical features.
- Groups similar songs.
- Recommends songs based on intrinsic characteristics.

3. OBJECTIVES

- Perform feature-driven analysis of songs.
- Identify latent clusters using unsupervised learning.
- Develop a real-time recommendation engine.
- Build an interactive visualization dashboard.
- Ensure scalability and extensibility.

4. DATASET DESCRIPTION

The system integrates multiple datasets to improve diversity and robustness.

4.1 Dataset Sources

Dataset	Description
Spotify Wrapped Top 50 Songs 2025	Popular songs in 2025
Spotify Wrapped Top 50 Artists 2025	Trending artists
Spotify All-Time Top 100 Songs	Historical hits

4.2 Feature Description

Feature	Meaning
Danceability	Suitability for dancing
Energy	Intensity and activity level
Valence	Positivity or happiness of the track
Acousticness	Likelihood of being acoustic

5. FEATURE SCALING

Distance-based algorithms require normalized inputs.

Technique Used -

MinMax Scaling :-

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- Prevents feature dominance.
- Improves clustering stability.
- Ensures fair similarity computation.

6. DATA PREPROCESSING

Data preprocessing ensures data consistency and quality.

Steps -

i) Dataset Merging -

```
df = pd.concat([df1, df2, df3], ignore_index=True)
```

ii) Column Standardization -

- Renamed inconsistent column labels.
- Ensured uniform schema.

iii) Duplicate Removal -

```
df = df.loc[:, ~df.columns.duplicated()]
```

iv) Missing Value Handling -

- Used median imputation to preserve distribution -

```
df[FEATURE_COLS] = df[FEATURE_COLS].fillna(df[FEATURE_COLS].median())
```

7. CLUSTERING MODEL (KMEANS)

Clustering is used to group songs based on feature similarity.

Algorithm - K-Means Clustering.

Working -

1. Initialize K centroids.
2. Assign points to nearest centroid.
3. Update centroids.
4. Repeat until convergence.

Why KMeans?

- Efficient for large datasets.
- Works well with numeric features.
- Easy to interpret.

Mathematical Objective -

Minimize :- $$\sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

Observations -

- Clusters represent music styles or moods.
- Example clusters -
  - High energy + high danceability → Party songs.
  - Low energy + high acousticness → Calm songs.

8. RECOMMENDATION SYSTEM

Approach - Content-Based Filtering.

Uses feature similarity instead of user behavior.

Similarity Metric -

Cosine Similarity -

$$\cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

Why Cosine Similarity?

- Measures orientation, not magnitude.
- Works well on normalized data.
- Captures feature alignment.

Workflow -

1. Convert songs into feature vectors.
2. Compute similarity matrix.
3. Rank songs based on similarity.
4. Return Top-N recommendations.

Example -

If input song = "Song A" -

- Compute similarity with all songs.
- Sort descending.
- Return top matches.

Strengths -

- No cold-start for items.
- Fast retrieval.
- Interpretable.

9. EXPLORATORY DATA ANALYSIS (EDA)

EDA helps uncover patterns and relationships.

Key Insights -

- Strong correlation between energy and danceability.
- Valence reflects emotional tone.
- Acousticness inversely related to energy.

Visualizations -

- Scatter plots (Energy vs Valence).
- Feature distributions.
- Radar charts.
- Cluster visualizations.

10. FRONTEND (STREAMLIT DASHBOARD)

The application is deployed using Streamlit.

Features -

- Interactive UI.
- Real-time recommendations.
- Dynamic clustering control.
- Visual analytics.

User Flow -

1. User opens dashboard.
2. Explores charts.
3. Inputs song name.
4. Receives recommendations.

11. CHALLENGES FACED

i) Data Inconsistency.

- Solved via standardization.

ii) Missing Values.

- Handled using median imputation.

iii) Feature Imbalance.

- Solved via normalization.

12. FUTURE SCOPE

Immediate Improvements -

- Mood-based filtering.
- PCA visualization.
- UI enhancements.

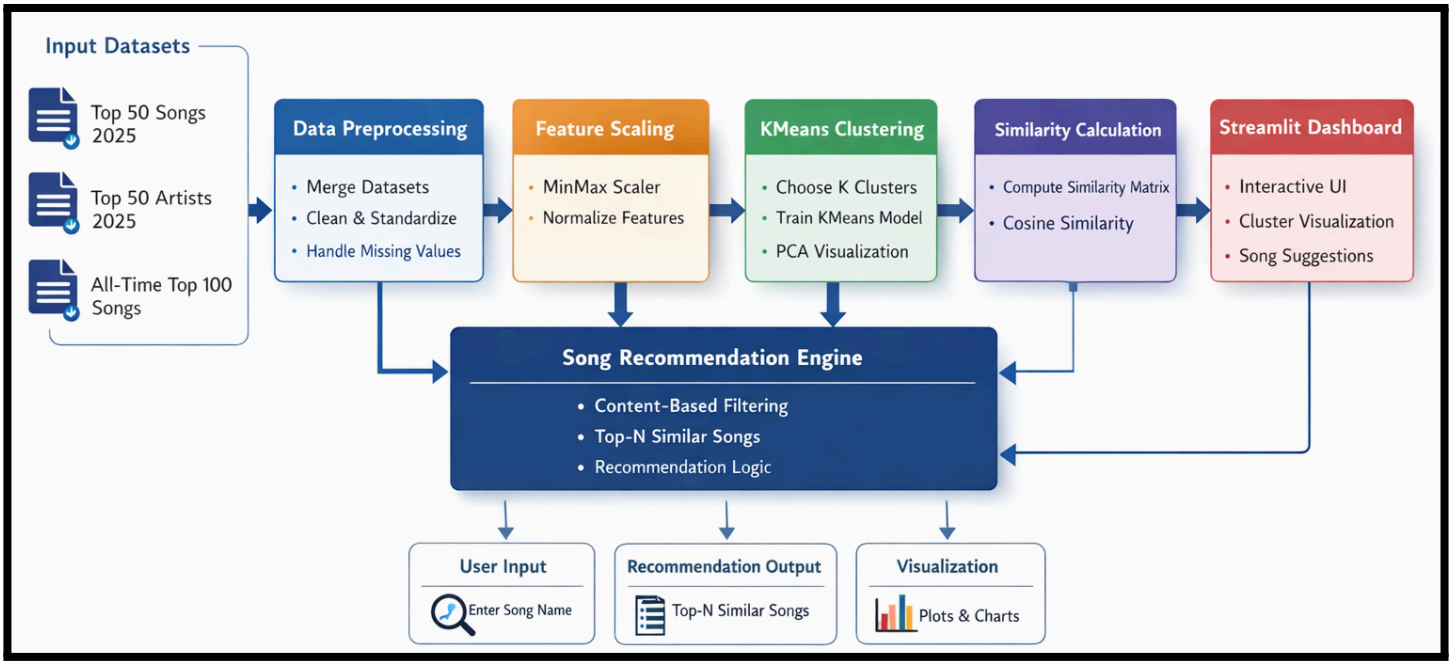
Advanced Enhancements -

- Spotify API integration.
- Hybrid recommendation system.
- Deep learning embeddings.
- Real-time streaming pipeline.

13. LIMITATIONS

- No user-personalized recommendations.
- No real-time streaming data.
- No evaluation metrics (precision/recall).

14. SYSTEM ARCHITECTURE



15. KEY LEARNINGS

- End-to-end ML pipeline design.
- Importance of feature engineering.
- Practical use of clustering.
- Building interactive ML applications.
- Visualization for storytelling.

16. PROJECT IMPACT

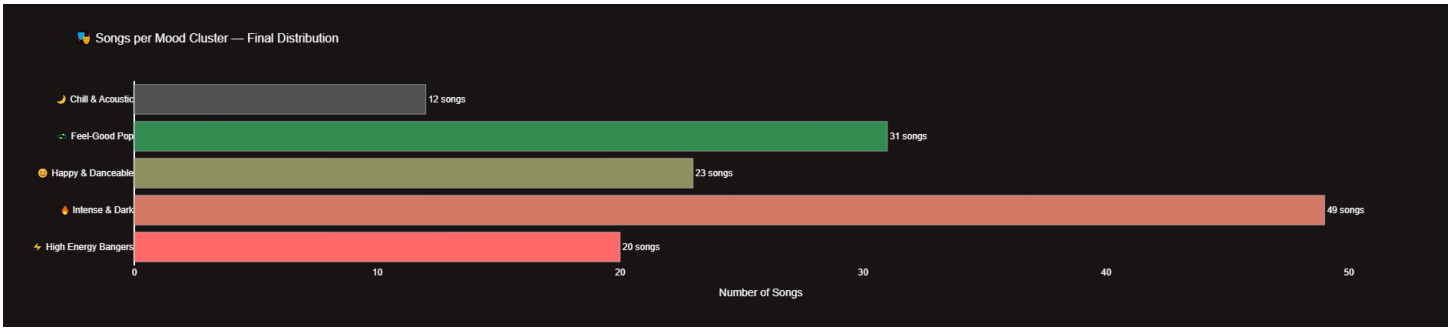
- Demonstrates industry-relevant ML skills.
- Strong portfolio project for -
    - Data Analyst roles.
    - ML Engineer roles.
    - Software Developer roles.

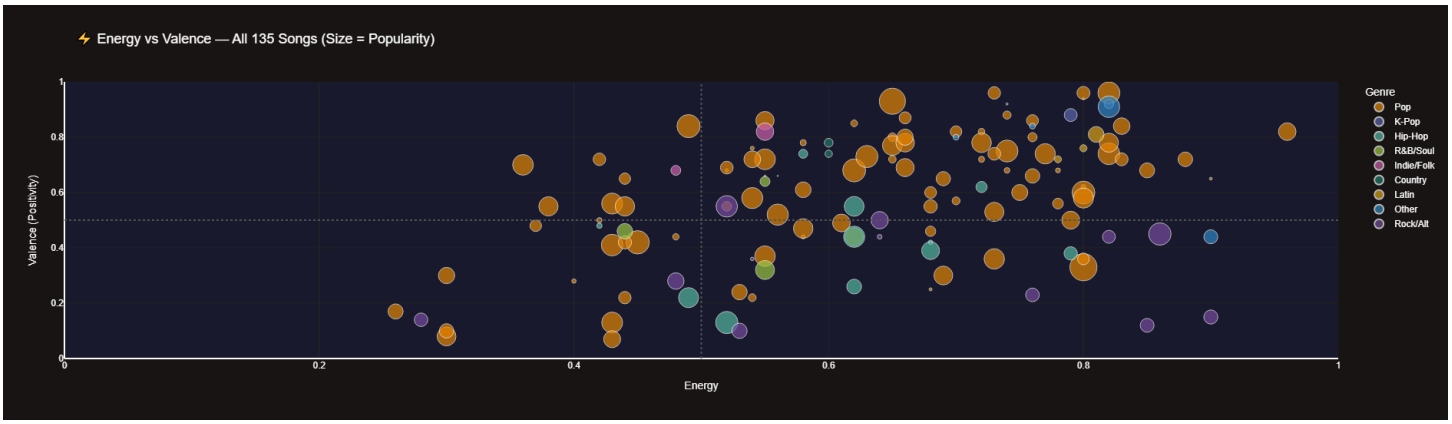
17. CONCLUSION

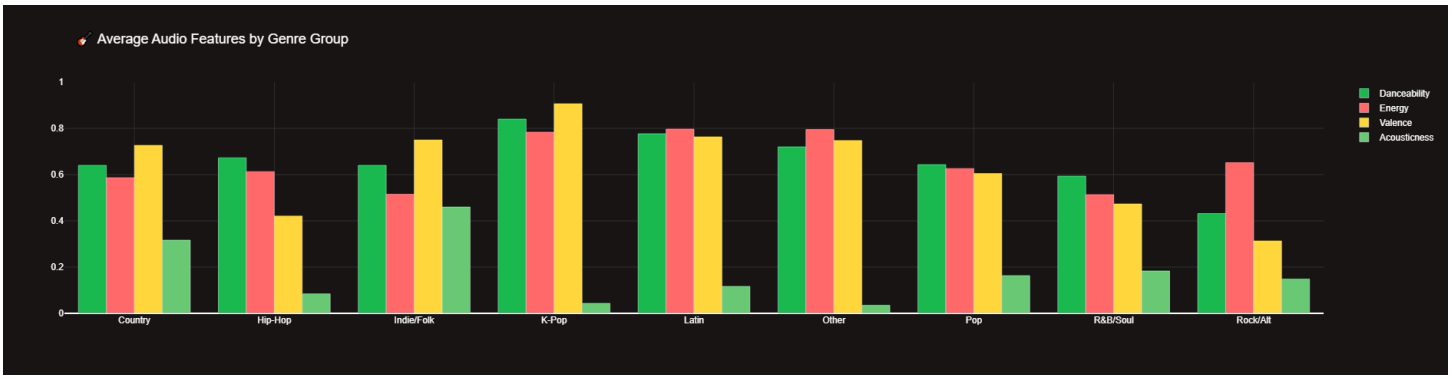
This project demonstrates a complete data science lifecycle, from raw data processing to deployment. It effectively combines machine learning, data visualization, and full-stack development to deliver a scalable recommendation system.

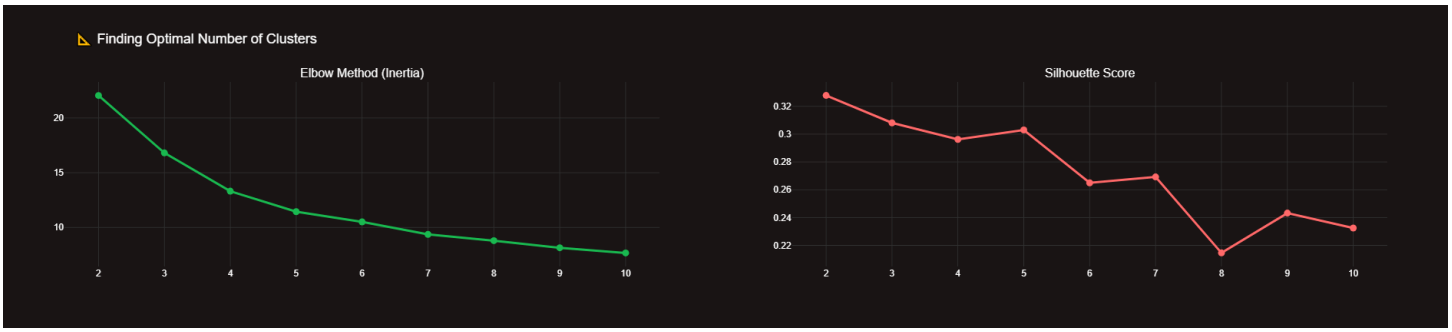
The system highlights how feature-driven similarity can power recommendation engines and lays the foundation for future enhancements toward industry-grade AI systems.

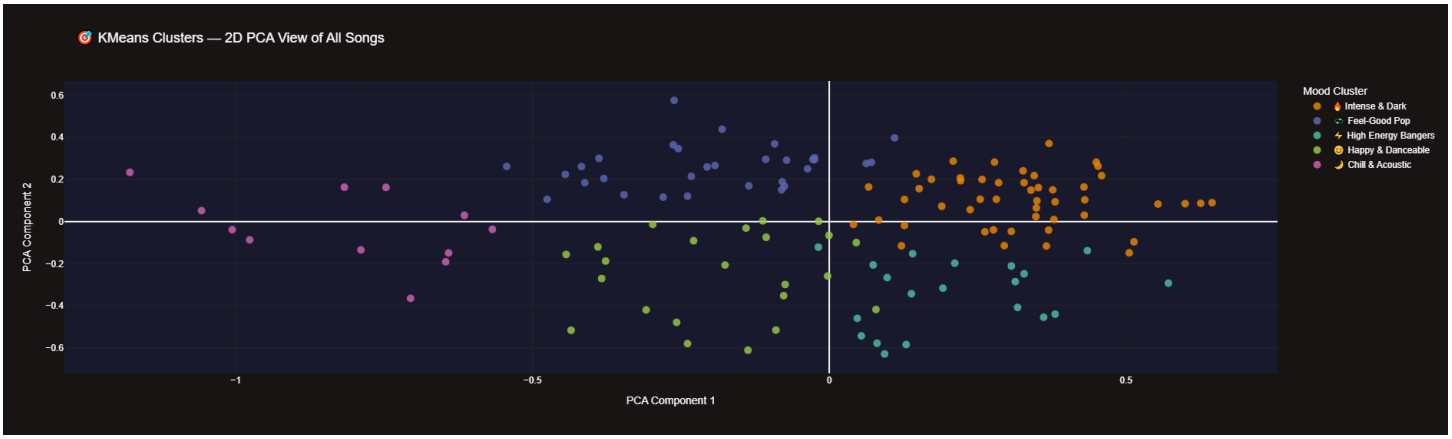
VISUALIZATIONS





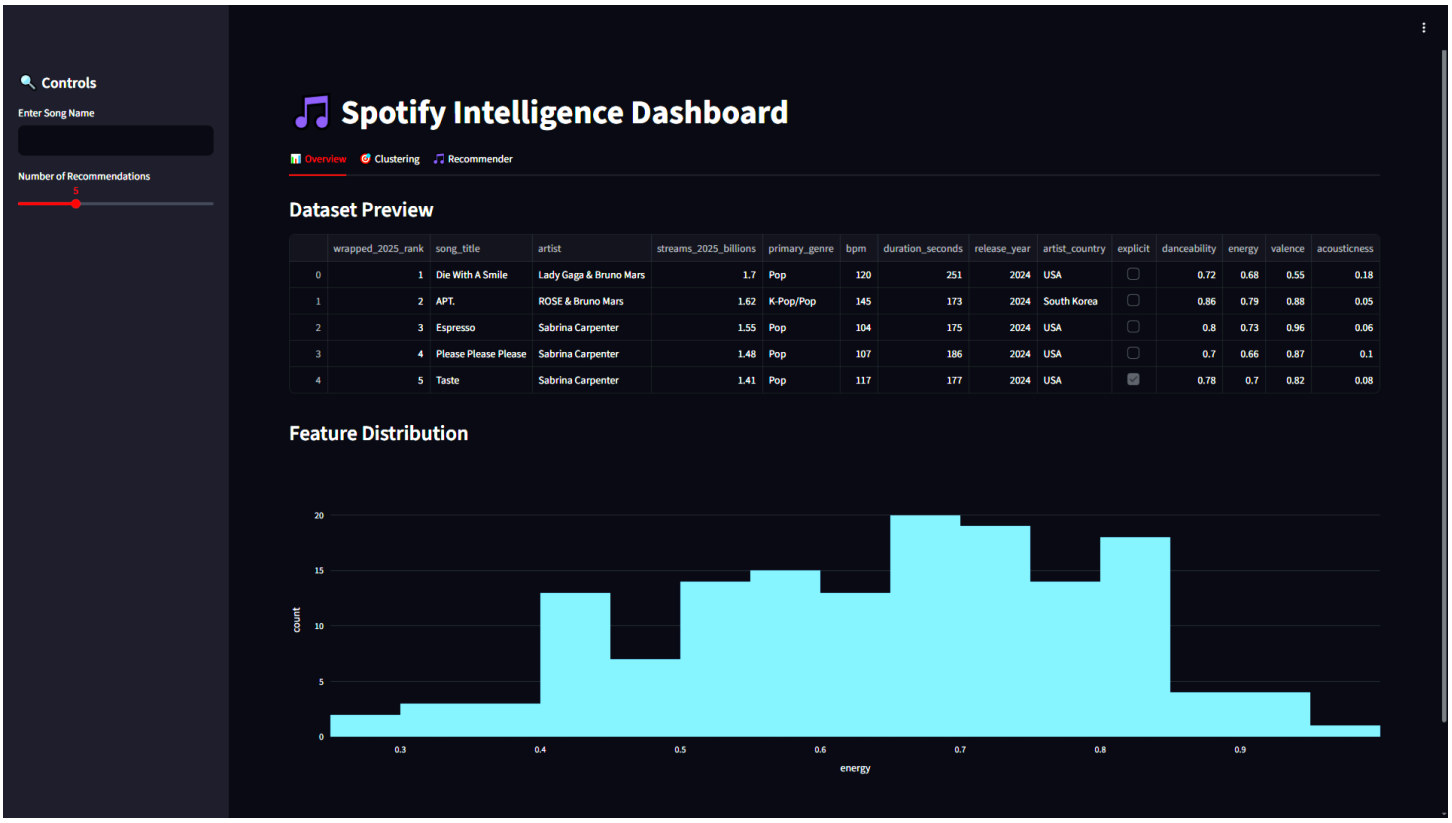




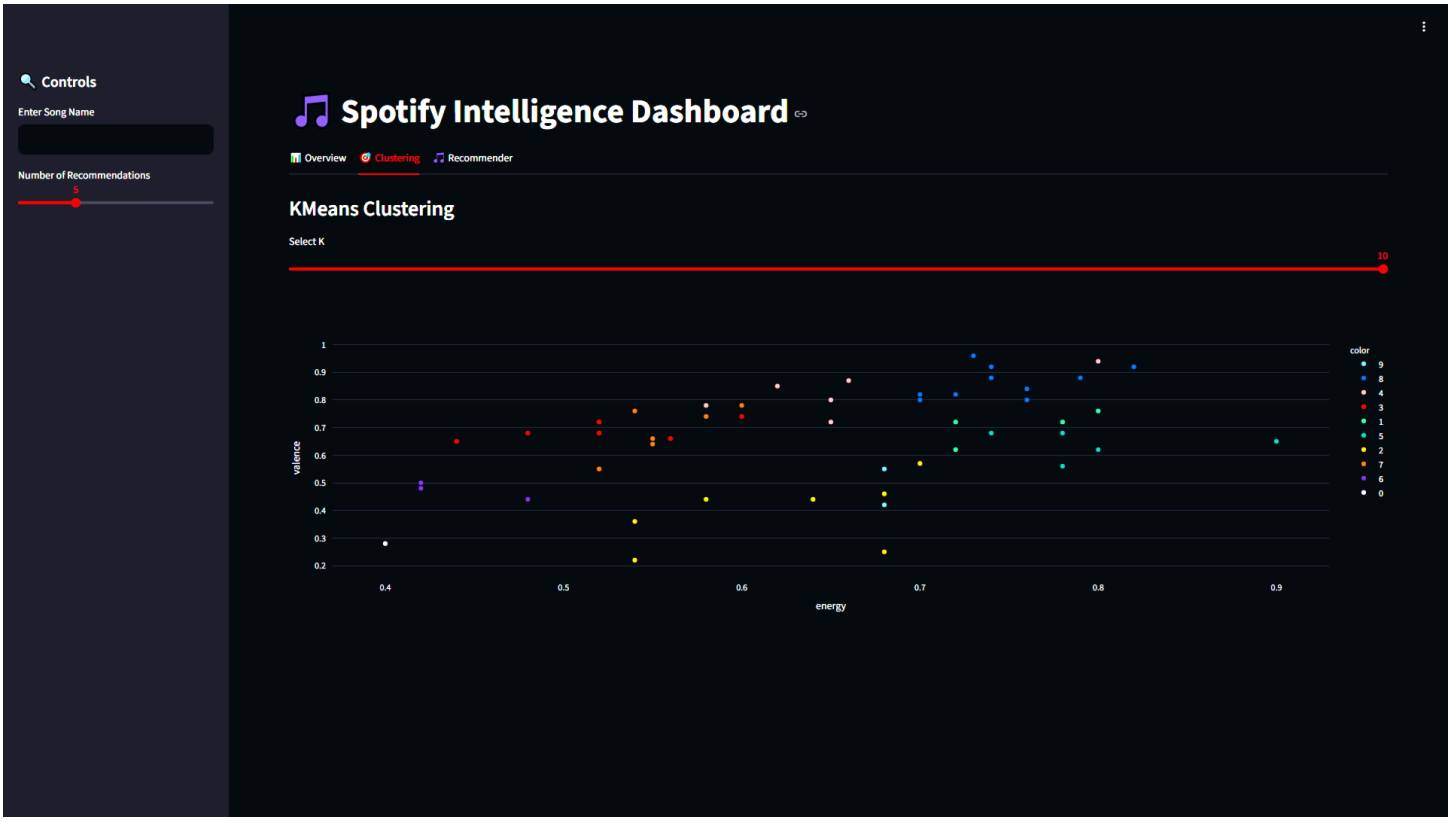


STREAMLIT SCREENSHOTS

GitHub Link - https://github.com/iamhriturajsaha/SPOTIFY-DATA-ANALYTICS-AND-SONG-RECOMMENDATION-ENGINE



	wrapped_2025_rank	song_title	artist	streams_2025_billions	primary_genre	bpm	duration_seconds	release_year	artist_country	explicit	danceability	energy	valence	acousticness
0	1	Die With A Smile	Lady Gaga & Bruno Mars	1.7	Pop	120	251	2024	USA	☐	0.72	0.68	0.55	0.18
1	2	APT.	ROSE & Bruno Mars	1.62	K-Pop/Pop	145	173	2024	South Korea	☐	0.86	0.79	0.88	0.05
2	3	Espresso	Sabrina Carpenter	1.55	Pop	104	175	2024	USA	☐	0.8	0.73	0.96	0.06
3	4	Please Please Please	Sabrina Carpenter	1.48	Pop	107	186	2024	USA	☐	0.7	0.66	0.87	0.1
4	5	Taste	Sabrina Carpenter	1.41	Pop	117	177	2024	USA	☑	0.78	0.7	0.82	0.08



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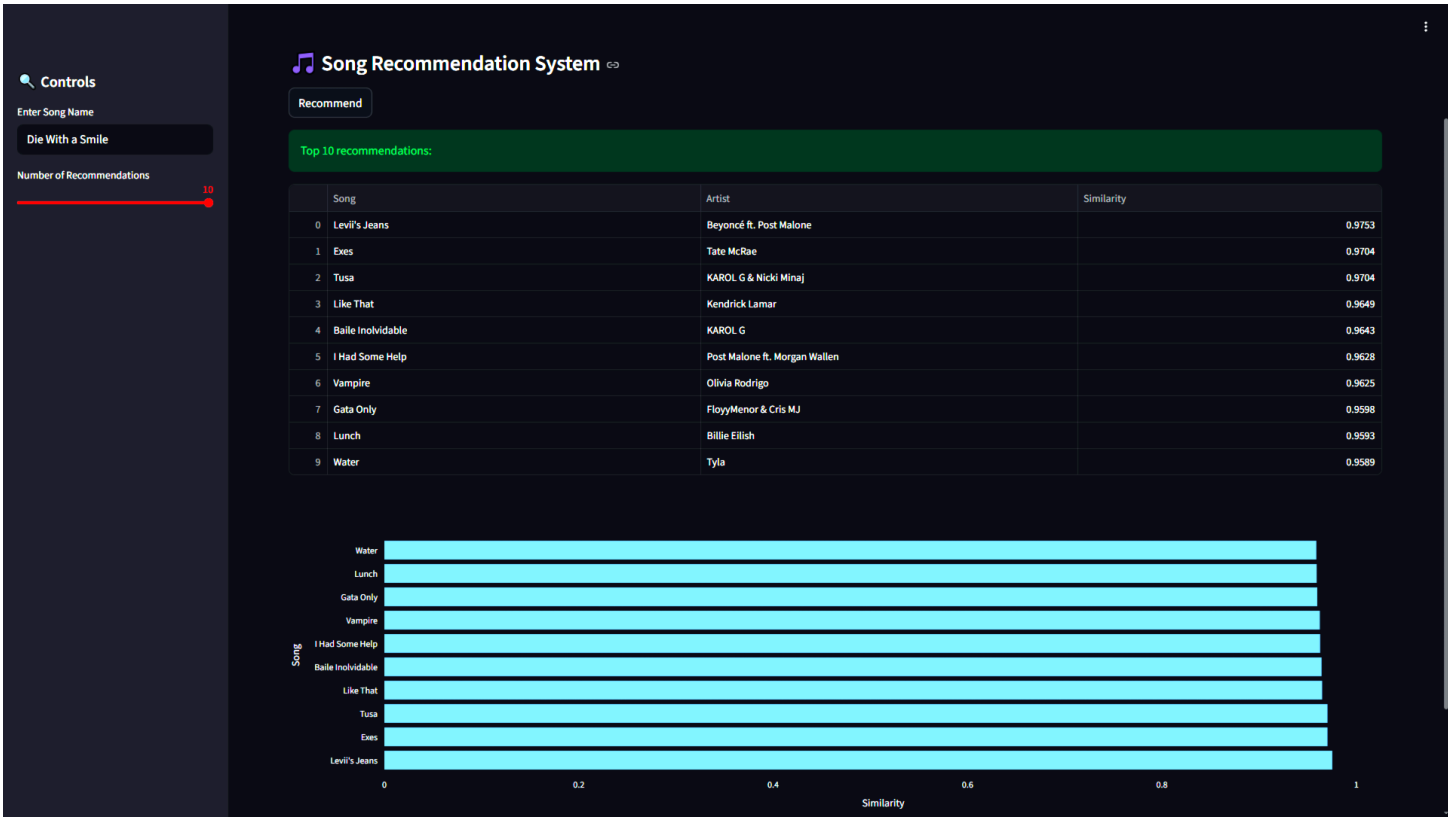

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